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Structural characterization and antioxidant activity of a polysaccharide from the fruiting bodies of cultured *Cordyceps militaris*

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Abstract

The water-soluble crude polysaccharides were obtained from the fruiting bodies of cultured *Cordyceps militaris* by hot water extraction followed by ethanol precipitation. The polysaccharides were successively purified by chromatography on DEAE–cellulose-52 and Sephacryl S-100 HR columns, giving main three polysaccharide fractions termed P50-1, P70-1, and P70-2. Structural features of P70-1 were investigated by a combination of chemical and instrumental analysis, such as partial acid hydrolysis, methylation analysis, periodate oxidation – Smith degradation, GC–MS, ¹³C NMR, HPAEC–PAD, and FT–IR. The results indicated that P70-1 has a backbone of (1 → 6)-linked β-D-mannopyranosyl residues, which occasionally branches at O-3. The branches were mainly composed of (1 → 4)-linked α-D-glucopyranosyl and (1 → 6)-linked β-D-galactopyranosyl residues, and terminated with β-D-galactopyranosyl residues and α-D-glucopyranosyl residues. In the in vitro antioxidant assay, P70-1 was found to possess hydroxyl radical-scavenging activity with an IC₅₀ value of 0.548 mg/ml.

Introduction

Interest in the structure of fungal extracellular polymers has increased since several physiological

roles were shown, in fungi morphogenesis, in their association with hosts and by their mobilization as food reserve (Krcmar, Novotny, Marais, & Joseleau, 1999). Many natural polysaccharides and polysaccharide–protein complexes were isolated from fungi and were used as a source of therapeutic agents (Carbonero et al., 2006, Mechacanon et al., 2005, Yu et al., 2004) during the past years.

Cordyceps militaris is known as the Chinese rare caterpillar fungus, and has similar pharmacological activities to the well known Chinese traditional medicine *Cordyceps sinensis* (Gai et al., 2004, Zheng and Cai, 2004). It is commonly used in the Orient to replenish the kidney and soothe the lung for the treatment of hyposexuality, hyperglycemia, hyperlipidemia, renal dysfunction, and liver disease (Won and Park, 2005, Yu et al., 2004). Recently, several studies have demonstrated that the extracts of *C. militaris* have multiple pharmacological actions such as anti-inflammatory (Won & Park, 2005), improvement of insulin resistance and insulin secretion (Choi, Park, Choi, Jun, & Park, 2004). The antioxidant activity of *C. militaris* is even stronger than *C. sinensis* and *Cordyceps kyushuensis* (Chen, Luo, Li, Sun, & Zhang, 2004). Because natural *C. militaris* is rare and expensive, many scientists have examined its life cycle with the aim of developing techniques for the isolating fermentable strains. Several strains have been isolated from natural *C. militaris* and manufactured in high quality by fungus-cultivation technology (Chai et al., 2003, Ling et al., 2003). The products from cultured *C. militaris* have shown similar pharmacological efficacy compared with natural *C. militaris* (Wang, 2002). Cultivated fruiting bodies of *C. militaris* were commonly sold as drug materials and health food products in China and South East Asia (Li et al., 2006, Zheng and Cai, 2004).

In the last few years, the structures of several polysaccharides isolated from *Cordyceps* spp. were reported (Mechacanon et al., 2005, Wu et al., 2006, Xiao et al., 2006). Three polysaccharides (CPS-1, CPS-2, and CPS-3) were isolated from *C. militaris* by our research group recently (Yu et al., 2004, Yu et al., 2004). The sugar composition analysis showed that CPS-1 was composed mainly of Rha, Xyl, Man, Glc, and Gla in a molar ratio of 1:6.43:25.6:16.0:13.8, CPS-2 consisted of Rha, Glc, and Gla in a molar ratio of 1:4.46:2.43 and CPS-3 was a homogeneous polysaccharide, which was composed of D-glucose with a major linkage form of a α -D-glucose. Although, the structural investigations of three polysaccharides shed light on the monosaccharide compositions of polysaccharides from *C. militaris*, isolation, purification, and activities' determination of more fine polysaccharides, especially detailed studies of the structures, were lacking.

Therefore, the polysaccharides were further fractionated from *C. militaris* and several distinctive polysaccharides were obtained. Among them, P70-1 exerted inhibitory activity towards hydroxyl radicals with an IC_{50} value of 0.548 mg/ml. So, the aim of this research was to investigate the complete structure of P70-1.

Section snippets

Material

Cultured *C. militaris* was obtained from Shenyang Zhongtian Bioengineering Corporation, Shenyang, China. The material (No. 97-08-0005) was identified by Professor R.M. Yu, College of Pharmacy, Jinan University, China.

Ascorbic acid (Vitamin C, Vc), hydrogen peroxide (H_2O_2), ferrous sulfate ($FeSO_4$), and brilliant green were purchased from Shanghai Chemical Reagent Company, Shanghai, China. All other reagents were obtained from the Sigma Chemical Co. All reagents were of analytical grade. ...

General methods

The total ...

Isolation, purification, and composition of polysaccharides

CP50 and CP70, two crude polysaccharides, were obtained from cultured *C. militaris* by hot water extraction followed by ethanol precipitation, with yields of 3% and 4%, respectively. After fractionation on DEAE-cellulose 52 and Sephacryl S-100 HR column, P70-1 (250 mg), P70-2 (30 mg) and P50-1 (40 mg) were obtained from the NaCl elution. The homogeneity of the three polysaccharides was elucidated by the following tests. They were all eluted as a single peak from gel-filtration chromatography on ...

Conclusion

The results of this paper showed that the water-soluble polysaccharide P70-1 of cultured *C. militaris* is a heteropolysaccharide and is occasionally branched. The fundamental information obtained from this work is beneficial to the interpretation in the relationship of the polysaccharide structure and its biological functions. The primary results of the in vitro antioxidant assay for the first time demonstrated that P70-1 possesses the hydroxyl radical-scavenging activity, which provides the ...

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